CANDIDATE

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TEST

A19055S5 Core Empirical Research Methods Part B 2024/25

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Section 1

Question	Question title	Question type
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Selected Paper (and motivation)

Chen, Y., & Zhong, S. (2025). People Are More Moral in Uncertain Environments. Econometrica, 93(2), 439-462. https://doi.org/10.3982/ECTA20574

The original replication files are here. The raw data set contains outcomes and responses recorded from the experiments, as described in sections below.

Summary of the Paper

Recent research shows that anxiety and complexity challenge standard models of decision-making under uncertainty, which assumes agents rationally weigh all possible outcomes according to their probabilities. The anxiety view says uncertainty can evoke feelings of fear, anxiety and stress; to offset these feelings, people act to gain a sense of control. This explanation aligns with beliefs such as moralistic gods or karma, in which moral behaviour brings favourable outcomes. The complexity view says people cannot predict every possible outcome due to the extreme complexity of the real world.

This paper provides robust empirical findings about facing uncertainties, people exhibit moral behaviour than the certain cases from a modified Dice Game. They show that this pattern is weakened when the moral implications of the behaviour are reduced or when the uncertainty pertains to others rather than oneself. The paper further contributes by proposing theoretical frameworks, including magical and quasi-magical thinking, to explain this phenomenon, suggesting that individuals act as if their moral behaviour can influence the resolution of uncertainty.

	Experiment	# Subject	Duration (mins)	Outcome Var	mean	sd	Average Payoffs
1	dice_game	107.00	18.24	$\mathbf{1}\{+4\}$	0.64	0.48	RMB44.06
2	direct_choice	102.00	15.21	$\mathbf{1}\{+4\}$	0.94	0.24	RMB42.73
3	second_party	107.00	26.62	$\mathbf{1}\{+4\}$	0.56	0.50	RMB36.91
4	dice_game_loss	305.00	17.68	$\mathbf{1}\{-4\}$	0.08	0.27	RMB43.45
5	ex_ante_resolution	191.00	14.23	$\mathbf{1}\{+2\}$	0.35	0.48	SGD19.77
6	dictator_game	148.00	13.69	$\mathbf{1}\{\text{share}\}$	0.20	0.40	SGD20.37
7	second_party_information	108.00	25.13	$\mathbf{1}\{+4\}$	0.63	0.48	RMB36.10

Note: Information about "Ave Payoffs" were manually inserted by the authors and not calculated from the replication data.

The experiment design (and key findings)

The main experiment builds on the dice game by Fischbacher and Follmi-Heusi (2013). First, subjects receive a lottery $(h, p; l)$, where p is the chance of winning a high payoff h; otherwise, they get a lower payoff l. Next, each subject mentally picks a number from 1 to 6. They are then told which number would give a bonus payoff of $a = 4$. After learning this, subjects report their chosen number. Lying cannot be detected at the individual level, but it is deemed as present at the aggregate level if the number associated with the bonus is reported more often than $\frac{1}{6}$ (by the LLN) of the time.

If a subject reports the number associated with the bonus, they play the lottery $(h+a, p; l+a)$. Otherwise, they play the lottery $(h, p; l)$.

The experiment puts each subject through 21 rounds. Payoffs such as $(40, p; 0)$, $(30, p; 10)$, and $(22, p; 18)$. Winning probabilities also vary: $p \in \{0, \frac{1}{6}, \frac{2}{6}, \frac{3}{6}, \frac{4}{6}, \frac{5}{6}, 1\}$.

In the result, the proportion of reporting the one with additional payoff is 59.1% under $p \in (0, 1)$, and a higher >74% under certainty outcomes. And this dispersion is amplified by increasing the spread between high payoff and low payoff.

To investigate underlying mechanisms, two supplementary experiments were conducted.

Direct Choice: In this version, subjects are first told to choose one out of six boxes without reporting their choices. Then they are told which box gives the extra payoff, and to open the box they have chosen. Fewer people choose the highest payoff when the outcome is uncertain than when it is certain. The drop in moral choices is also smaller here than in the Dice Game.

Second Party Game: Player A is given the lottery $(h, p; l)$, but does not make any decision. Player B receives a fixed payoff and need to decide whether to act honestly for an additional payoff for Player B herself.

Other Robust Experiments

The paper also conducted three experiments for robustness check.

Dice Game Loss Experiment: Subjects are less likely to lie to avoid losses under uncertainty than under certainty, consistent with the main finding.

Ex Ante Resolution Experiment: Even when uncertainty is resolved ex-ante but kept unknown (to check if is there any difference between treating a not revealed result and a revealed result but the subject does not have the information), subjects are less likely to lie under uncertainty than certainty.

Dictator Game Experiment: In the domain of other-regarding behaviour, subjects were more likely to share with others in uncertain environments than in certain ones.

Regression Analysis

The paper's main empirical strategy is an OLS regression at the individual-experiment level. The dependent variable is a measure of dishonesty: it takes the value 1 when the subject reports the box or the number associated with the bonus and 0 otherwise.

The regressions in Table 2 are built on the following equation but they have different samples with respect to different experimental settings.

$$Y_i = \beta_0 + \beta_1 \mathbf{1}\{h\}_i + \beta_2 \mathbf{1}\{l\}_i + \gamma X_i + \epsilon_i$$

where Y_i is the indicator of reporting the number associated with the bonus for subject i, $\mathbf{1}\{h\}_i$ is an indicator that takes the value 1 if the subject faces a certain high payoff, $\mathbf{1}\{l\}_i$ is an indicator that takes the value 1 if the subject faces a certain low payoff, and X_i is a vector of control variables including individual fixed effects, decision duration, and sequence of decisions.

This estimation aims to quantify the causal effect of environmental certainty (both high payoff and low payoff certainty) on an individual's likelihood of engaging in dishonest behaviour and to get the interpretation of how uncertainty leads to less 'add4' reporting rate as a proxy for effect on moral behaviour. The coefficients on $\mathbf{1}\{h\}$ and $\mathbf{1}\{l\}$ reflect how much more or less often the subject reports the number associated with the bonus in certain high payoff and certain low payoff setups. For example, if the coefficient on $\mathbf{1}\{l\}$ is 0.19, the subject is on average 19 percentage points less likely to report the number with the bonus in an uncertain lottery than in a certain low payoff one.

The R code successfully replicates the results in Table 2 of the paper. When the difference between high payoff and low payoff is large, see columns (40, p; 0) and (30, p; 10); the uncertainty effect is more sizeable than the (22, p; 18) case: the coefficients in the first two

cases are more than 5 times of the last ones. The Panel B checked the uncertainty effect by picking subsamples by the winning probabilities and it confirms the panel a) in Fig. 1 displayed in the 'Code' part.

The paper also conducted a robustness check by using a probit model, which gives similar results. Based on my code, the paper is indeed replicable. Although there are small differences between the coefficients in Table_A_3 in my replication and the paper, it does not affect the robustness of the results and the findings.

Since the datasets are collected from experiments, if there is no selection bias, the OLS regression with cluster is sufficient to get the causal effect. Therefore, conducting the same experiment on people with different cultures may verify if this result is affected by the cultural and education background. Or if having social experience would change the results. For example, people from some cultural backgrounds which are strongly influenced by collectivist social norms might behave differently from those from individualist cultures.

Code and Figures

Data cleaning in the original replication package was intensive, comprising seven cleaning scripts, each exceeding 150 lines of code. I tried to write a generalised R function for data cleaning while adapting data from multiple experiments; thus, my data cleaning implementation differs substantially from the original package's structure.

The generalised function successfully processed data for five of the seven experiments. The remaining two experiments exhibited significant structural differences in their experimental design. To ensure comprehensive data availability for the replication, I loaded two cleaned datasets in the replication package and 5 datasets I cleaned with my function. As part of this replication, I reproduced **four** key figures and tables successfully, as compensation for the incomplete data cleaning. Also, I attached my data cleaning part in the appendix section below.

```
<environment: R_GlobalEnv>
```

Table A.1: Summary Statistics

```r
# Create Table A.1
# A function to obtain needed statistics

data_list <- list(
  dice_game = dice_game_cleaned,
  direct_choice = direct_choice_cleaned,
```

```r
  second_party = second_party,
  dice_game_loss = dice_game_loss_cleaned,
  ex_ante_resolution = ex_ante_resolution_cleaned,
  dictator_game = dictator_game_cleaned,
  second_party_information = second_party_information
)

outcome_var <- c("add4", "add4", "add4", "deduct4", "add2", "share", "add4")

summary_statistics <- function(dataset, outcome) {
  results <- dataset |>
  summarise(n_subj = max(subject, na.rm = T),
  duration_min = mean(duration_in_seconds / 60, na.rm = TRUE),
  mean_outcome = mean(.data[[outcome]], na.rm = TRUE),
  sd_outcome = sd(.data[[outcome]], na.rm = TRUE))

  results
}

# Obtain statistics and plot the table

tab1 <- map2(data_list, outcome_var, summary_statistics)
tab1_df <- bind_rows(tab1, .id = "Experiment")
tab1_df <- tab1_df |>
  mutate(average_payoffs = c('RMB44.06', 'RMB42.73', 'RMB36.91', 'RMB43.45',
                             'SGD19.77', 'SGD20.37', 'RMB36.10'),
         var = c('1{+4}', '1{+4}', '1{+4}', '1{-4}',
                 '1{+2}', '1{share}', '1{+4}')) |>
  relocate(var, .after = duration_min) |>
  rename('# Subject' = n_subj,
         'Duration (mins)' = duration_min,
         'Average Payoffs' = average_payoffs,
         "Outcome Var" = var)
tab1_df
```

```
# A tibble: 7 x 7
  Experiment `# Subject` `Duration (mins)` `Outcome Var` mean_outcome sd_outcome
  <chr>            <dbl>             <dbl> <chr>                <dbl>      <dbl>
1 dice_game          107              18.2 1{+4}               0.640       0.480
2 direct_ch~         102              15.2 1{+4}               0.940       0.237
3 second_pa~         107              26.6 1{+4}               0.559       0.497
4 dice_game~         305              17.7 1{-4}               0.0778      0.268
```

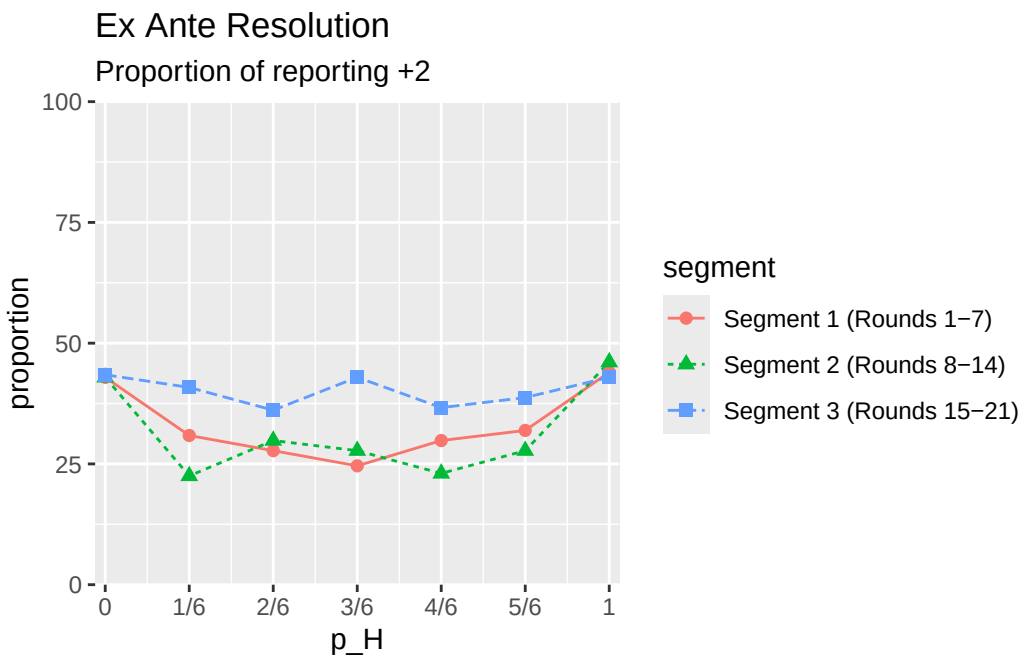
```
5 ex_ante_r~         191              14.2 1{+2}                0.349       0.477
6 dictator_~         148              13.7 1{share}             0.203       0.403
7 second_pa~         108              25.1 1{+4}                0.630       0.483
# i 1 more variable: `Average Payoffs` <chr>
```

Figure 1: Proportion of Reporting +4 Across Experiments

```r
x_breaks <- c(0, 1/6, 2/6, 0.5, 4/6, 5/6, 1)
x_labels <- c("0", "1/6", "2/6", "3/6", "4/6", "5/6", "1")
y_limits <- c(0, 100)

# Function to process and plot each dataset
create_content_plot <- function(data,
                                value_var_name,
                                title_text,
                                subtitle_text) {
  df_processed <- data |>
    group_by(round) |>
    summarise(
      value = mean(.data[[value_var_name]], na.rm = TRUE),
      p_H = mean(p_H, na.rm = TRUE),
      .groups = 'drop'
    ) |>
    mutate(
      proportion = value * 100,
      segment = case_when(
        round <= 7 ~ "Segment 1 (Rounds 1-7)",
        round > 7 & round <= 14 ~ "Segment 2 (Rounds 8-14)",
        round > 14 & round <= 21 ~ "Segment 3 (Rounds 15-21)",
        round > 21 & round <= 28 ~ "Segment 4 (Rounds 22-28)",
        round > 28 & round <= 35 ~ "Segment 5 (Rounds 29-35)",
        round > 35 & round <= 42 ~ "Segment 6 (Rounds 36-42)",
        TRUE ~ NA_character_
      )
    )
    ggplot(df_processed, aes(x = p_H, y = proportion, color = segment,
                             shape = segment, linetype = segment)) +
    geom_line() +
    geom_point(size = 2) +
    # X-axis customization
    scale_x_continuous(
```

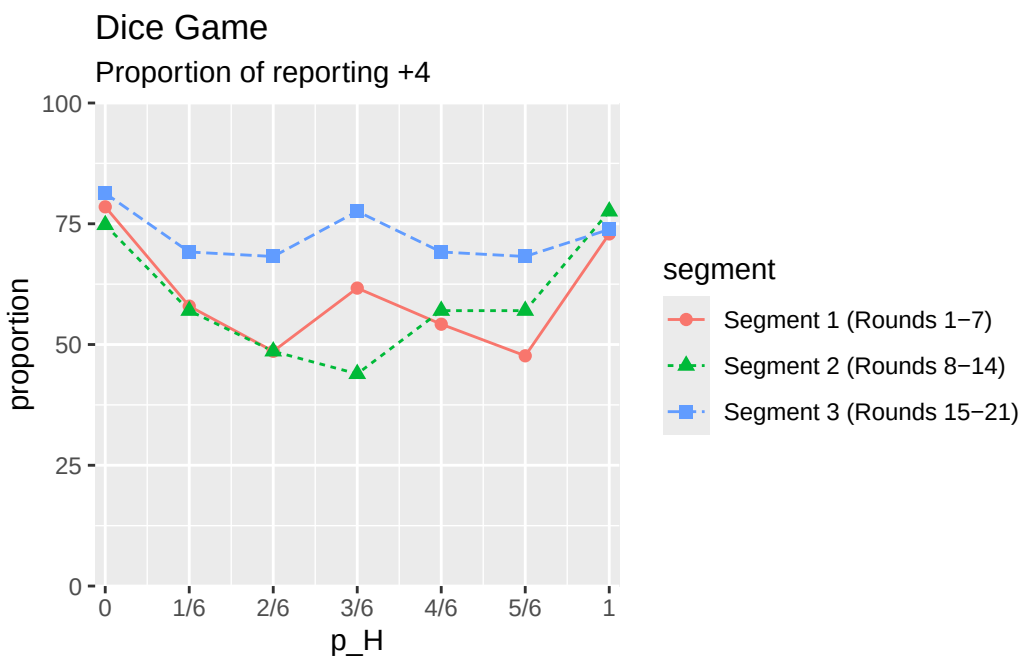
```r
      breaks = x_breaks,
      labels = x_labels,
      expand = expansion(mult = c(0.02, 0.02))
    ) +
    # Y-axis customization
    scale_y_continuous(
      limits = y_limits,
      expand = c(0, 0)
    ) +
    labs(
      title = title_text,
      subtitle = subtitle_text
    )
}

# 1. Dice Game Plot
plot_dice_game <- create_content_plot(
  data = dice_game_cleaned,
  value_var_name = "add4",
  title_text = "Dice Game",
  subtitle_text = "Proportion of reporting +4"
)


# 2. Direct Choice Plot
plot_direct_choice <- create_content_plot(
  data = direct_choice_cleaned,
  value_var_name = "add4",
  title_text = "Direct Choice",
  subtitle_text = "Proportion of reporting +4"
)

# 3. Second Party Plot
plot_second_party <- create_content_plot(
  data = second_party,
  value_var_name = "add4",
  title_text = "Second Party",
  subtitle_text = "Proportion of reporting +4"
)

# 4. Dice Game Loss Plot
plot_dice_game_loss <- create_content_plot(
```

```
  data = dice_game_loss_cleaned,
  value_var_name = "deduct4",
  title_text = "Dice Game Loss",
  subtitle_text = "Proportion of reporting -4"
)

# 5. Ex Ante Resolution Plot
plot_ex_ante_resolution <- create_content_plot(
  data = ex_ante_resolution_cleaned,
  value_var_name = "add2",
  title_text = "Ex Ante Resolution",
  subtitle_text = "Proportion of reporting +2"
)
print(plot_ex_ante_resolution)
```

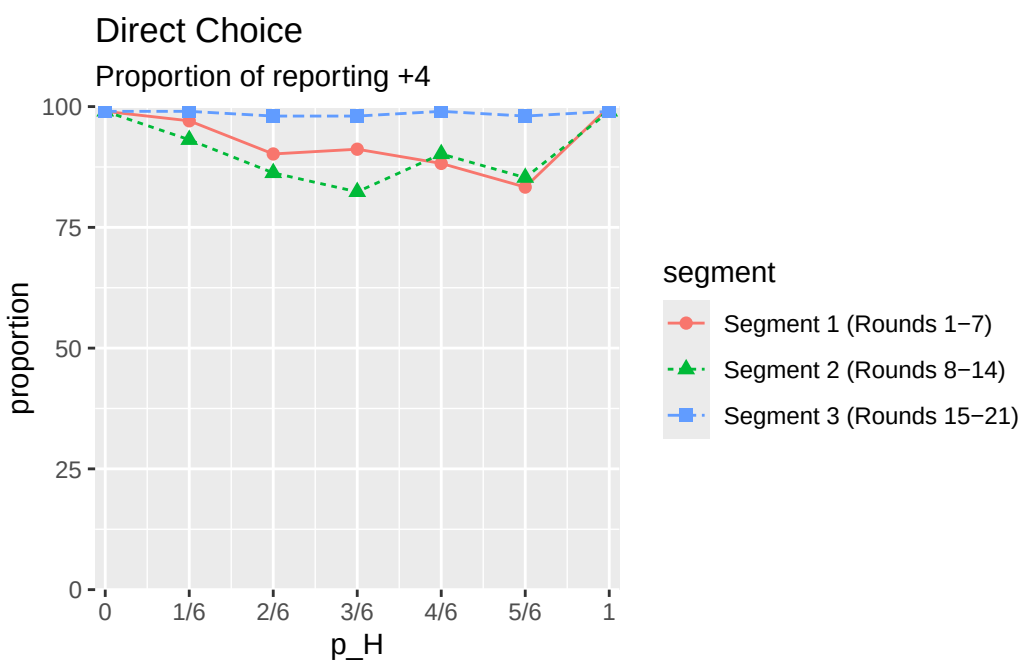



```
# 6. Dictator Game Plot
plot_dictator_game <- create_content_plot(
  data = dictator_game_cleaned,
  value_var_name = "share",
  title_text = "Dictator Game",
  subtitle_text = "Proportion of Sharing"
)
```

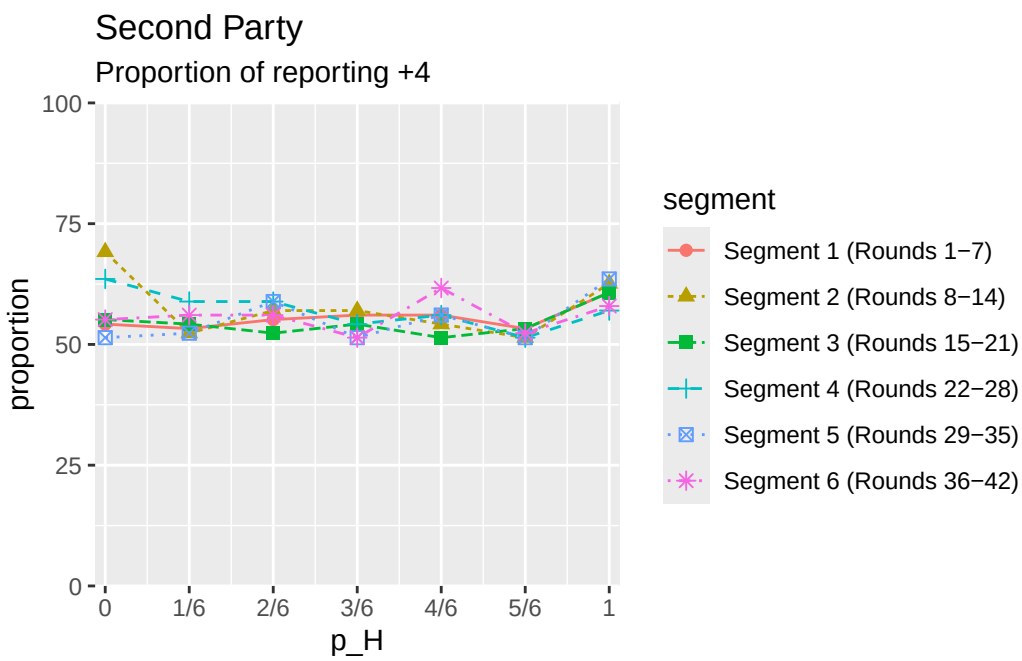
```
plot_dice_game
```



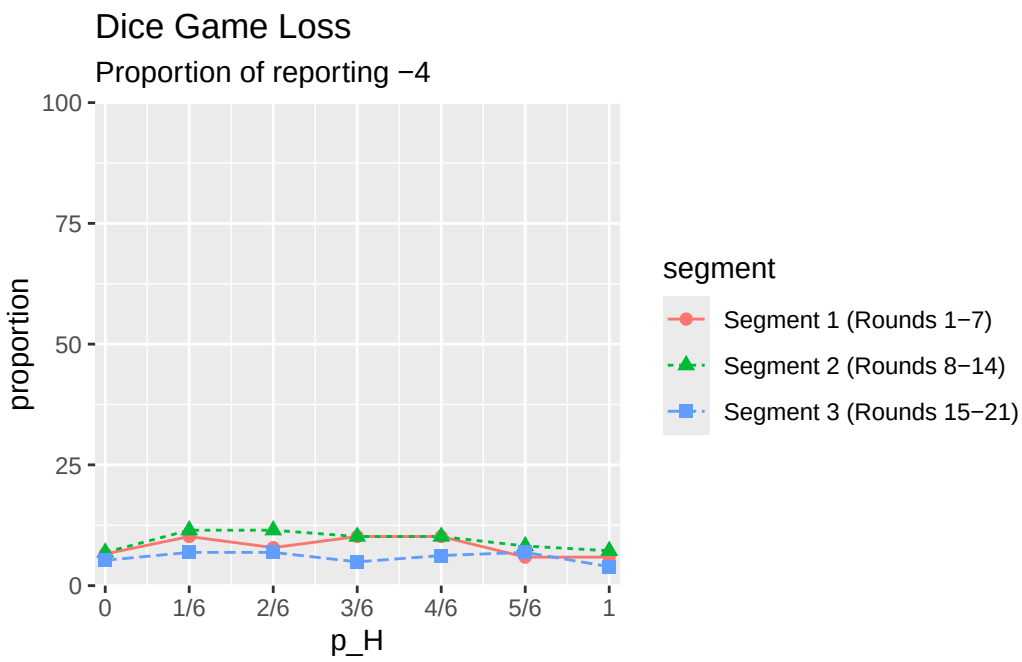

```
plot_direct_choice
```



```
plot_second_party
```




```
plot_dice_game_loss
```



```
plot_dictator_game
```

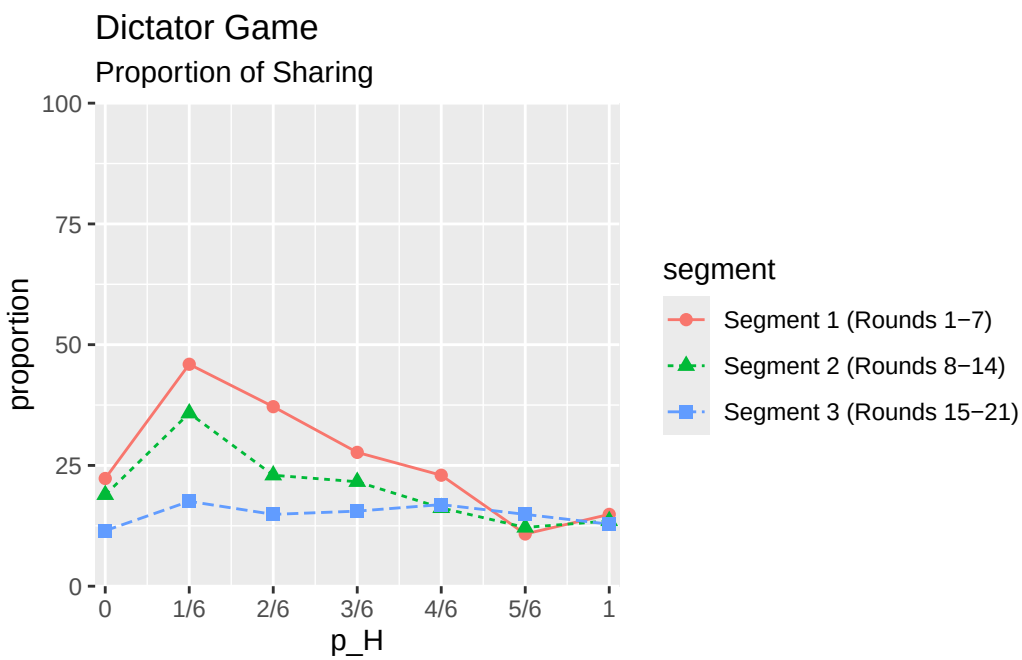


Table 2: Regression Results of Dice Game Experiment

```
# panel a
dice_game_cleaned <- dice_game_cleaned |>
  mutate(
    H = p_H == 1,
    L = p_H == 0,
    group_no = as.factor(group_no),
    subject = as.factor(subject)
  )
reg1a <- lm_robust(formula = add4 ~ H + L,
                   data = dice_game_cleaned, cluster = subject)

reg2a <- lm_robust(formula = add4 ~ H + L +
                     decision_time_task1_in_round +
                     decision_time_task2_in_round + order_of_round +
                     group_no + subject, data = dice_game_cleaned,
                   cluster = subject)
```

```r
reg3a <- lm_robust(formula = add4 ~ H + L +
                     decision_time_task1_in_round +
                     decision_time_task2_in_round + order_of_round +
                     subject,
                   data = dice_game_cleaned |> filter(group_no == 1),
                   cluster = subject)

reg4a <- lm_robust(formula = add4 ~ H + L +
                     decision_time_task1_in_round +
                     decision_time_task2_in_round + order_of_round +
                     subject,
                   data = dice_game_cleaned |> filter(group_no == 2),
                   cluster = subject)

reg5a <- lm_robust(formula = add4 ~ H + L +
                     decision_time_task1_in_round +
                     decision_time_task2_in_round + order_of_round +
                     subject,
                   data = dice_game_cleaned |> filter(group_no == 3),
                   cluster = subject)

# panel b

reg1b <- lm_robust(formula = add4 ~ H + L + decision_time_task1_in_round +
                     decision_time_task2_in_round + order_of_round +
                     group_no + subject,
                   data = dice_game_cleaned |>
                     filter(p_H == 1 | p_H == 0 | round == 2 |
                              round == 9  | round == 16),
                   cluster = subject,
                   se_type = "stata")

reg2b <- lm_robust(formula = add4 ~ H + L + decision_time_task1_in_round +
                     decision_time_task2_in_round + order_of_round +
                     group_no + subject,
                   data = dice_game_cleaned |>
                     filter(p_H == 1 | p_H == 0 | round == 3 |
                              round == 10  | round == 17),
                   cluster = subject,
                   se_type = "stata")

reg3b <- lm_robust(formula = add4 ~ H + L +
```

```
                      decision_time_task1_in_round +
                      decision_time_task2_in_round +
                      order_of_round + group_no + subject,
                    data = dice_game_cleaned |>
                      filter(p_H == 1 | p_H == 0 | round == 4 |
                               round == 11  | round == 18),
                    cluster = subject)
reg4b <- lm_robust(formula = add4 ~ H + L +
                      decision_time_task1_in_round +
                      decision_time_task2_in_round +
                      order_of_round + group_no + subject,
                    data = dice_game_cleaned |>
                      filter(p_H == 1 | p_H == 0 | round == 5 |
                               round == 12  | round == 19),
                    cluster = subject)
reg5b <- lm_robust(formula = add4 ~ H + L +
                      decision_time_task1_in_round +
                      decision_time_task2_in_round +
                      order_of_round + group_no + subject,
                    data = dice_game_cleaned |>
                      filter(p_H == 1 | p_H == 0 | round == 6 |
                               round == 13  | round == 20),
                    cluster = subject)
modelsummary(list(All = reg1a, All = reg2a, '40,p;0' = reg3a, '30,p;0' = reg4a, '22,p;18' = r
             gof_map = c('nobs', 'r.squared'), coef_omit = c(
               'subject|group_no|decision_time_task1_in_round|decision_time_task2_in_round')
             add_rows = data.frame(
               "term" = "Controls",
               "Model 1" = "N",
               "Model 2" = "Y",
               "Model 3" = "Y",
               "Model 4" = "Y",
               "Model 5" = "Y",
               check.names = FALSE
               ),
             title = 'Table 2: Regression Results of Dice Game Experiment (Panel A. Full samp
)
```

```
modelsummary(list('1/6' = reg1b, '2/6' = reg2b, '3/6' = reg3b,
                  '4/6' = reg4b, '5/6' = reg5b),
             gof_map = c('nobs', 'r.squared'), coef_omit = c(
               'subject|group_no|decision_time_task1_in_round
```

Table 1: Table 2: Regression Results of Dice Game Experiment (Panel A. Full sample and subsamples by payoff pairs)

	All	All	40,p;0	30,p;0	22,p;18
(Intercept)	0.591	0.299	0.385	0.232	0.381
	(0.029)	(0.028)	(0.028)	(0.046)	(0.046)
HTRUE	0.157	0.155	0.188	0.247	0.030
	(0.028)	(0.028)	(0.043)	(0.041)	(0.037)
LTRUE	0.191	0.190	0.242	0.218	0.108
	(0.025)	(0.025)	(0.041)	(0.040)	(0.036)
order_of_round		0.001	0.001	0.000	0.003
		(0.001)	(0.002)	(0.003)	(0.003)
Num.Obs.	2247	2247	749	749	749
R2	0.027	0.383	0.447	0.451	0.424
Controls	N	Y	Y	Y	Y

```
    |decision_time_task2_in_round'),
  add_rows = data.frame(
    "term" = "Controls",
    "Model 1" = "Y",
    "Model 2" = "Y",
    "Model 3" = "Y",
    "Model 4" = "Y",
    "Model 5" = "Y",
    check.names = FALSE
    ),
  title = 'Table 2: Regression Results of Dice Game Experiment
  (Panel B. Subsamples by winning probabilities)'
)
```

Table 2: Table 2: Regression Results of Dice Game Experiment (Panel B. Subsamples by winning probabilities)

	1/6	2/6	3/6	4/6	5/6
(Intercept)	0.120	0.180	0.357	0.330	0.140
	(0.046)	(0.050)	(0.050)	(0.045)	(0.046)
HTRUE	0.131	0.195	0.135	0.142	0.171
	(0.032)	(0.039)	(0.034)	(0.037)	(0.033)
LTRUE	0.166	0.230	0.170	0.176	0.206
	(0.031)	(0.033)	(0.032)	(0.036)	(0.031)
decision_time_task1_in_round	−0.008	−0.002	−0.010	−0.007	−0.004
	(0.006)	(0.006)	(0.007)	(0.005)	(0.007)
order_of_round	0.001	0.000	0.001	0.000	0.003
	(0.002)	(0.003)	(0.002)	(0.002)	(0.002)
Num.Obs.	963	963	963	963	963
R2	0.470	0.465	0.424	0.415	0.470
Controls	Y	Y	Y	Y	Y

TableA3: Robustness Check for Table 2

```
# Table A.3: Robustness check for table 2
reg1a_probit <- glm(formula = add4 ~ H + L,
                    data = dice_game_cleaned,
                    family = binomial(link = "probit"))
reg1a_clustered_sd <- vcovCL(reg1a_probit, cluster = ~ subject)
mea_reg1a <- margins(reg1a_probit, vcov = reg1a_clustered_sd, atmeans = TRUE)

reg2a_probit <- glm(formula = add4 ~ H + L +
                      decision_time_task1_in_round +
                      decision_time_task2_in_round + order_of_round +
                      group_no + subject,
                    data = dice_game_cleaned,
                    family = binomial(link = "probit"))
reg2a_clustered_sd <- vcovCL(reg2a_probit, cluster = ~ subject)
mea_reg2a <- margins(reg2a_probit, variables = c("H", "L"),
                      vcov = reg2a_clustered_sd, atmeans = TRUE)

reg3a_probit <- glm(formula = add4 ~ H + L +
```

```
                      decision_time_task1_in_round +
                      decision_time_task2_in_round + order_of_round +
                      subject,
                    data = dice_game_cleaned |> filter(group_no == 1),
                    family = binomial(link = "probit"))
reg3a_clustered_sd <- vcovCL(reg3a_probit, cluster = ~ subject)
mea_reg3a <- margins(reg3a_probit, variables = c("H", "L"),
                      vcov = reg3a_clustered_sd, atmeans = TRUE)

reg4a_probit <- glm(formula = add4 ~ H + L +
                      decision_time_task1_in_round +
                      decision_time_task2_in_round + order_of_round +
                      subject,
                    data = dice_game_cleaned |> filter(group_no == 2),
                    family = binomial(link = "probit"))
reg4a_clustered_sd <- vcovCL(reg4a_probit, cluster = ~ subject)
mea_reg4a <- margins(reg4a_probit, variables = c("H", "L"),
                      vcov = reg4a_clustered_sd, atmeans = TRUE)

reg5a_probit <- glm(formula = add4 ~ H + L +
                      decision_time_task1_in_round +
                      decision_time_task2_in_round + order_of_round +
                      subject,
                    data = dice_game_cleaned |> filter(group_no == 3),
                    family = binomial(link = "probit"))
reg5a_clustered_sd <- vcovCL(reg5a_probit, cluster = ~ subject)
mea_reg5a <- margins(reg5a_probit, variables = c("H", "L"),
                      vcov = reg5a_clustered_sd, atmeans = TRUE)

reg1b_probit <- glm(formula = add4 ~ H + L + decision_time_task1_in_round +
                      decision_time_task2_in_round + order_of_round +
                      group_no + subject,
                    data = dice_game_cleaned |>
                      filter(p_H == 1 | p_H == 0 | round == 2 |
                               round == 9  | round == 16),
                    family = binomial(link = "probit"))
```

```
Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
reg1b_clustered_sd <- vcovCL(reg1b_probit, cluster = ~ subject)
mea_reg1b <- margins(reg1b_probit, variables = c("H", "L"),
```

```
                      vcov = reg1b_clustered_sd, atmeans = TRUE)

reg2b_probit <- glm(formula = add4 ~ H + L + decision_time_task1_in_round +
                      decision_time_task2_in_round + order_of_round +
                      group_no + subject,
                    data = dice_game_cleaned |>
                      filter(p_H == 1 | p_H == 0 | round == 3 |
                               round == 10  | round == 17),
                    family = binomial(link = "probit"))
reg2b_clustered_sd <- vcovCL(reg2b_probit, cluster = ~ subject)
mea_reg2b <- margins(reg2b_probit, variables = c("H", "L"),
                      vcov = reg2b_clustered_sd, atmeans = TRUE)

reg3b_probit <- glm(formula = add4 ~ H + L +
                      decision_time_task1_in_round +
                      decision_time_task2_in_round +
                      order_of_round + group_no + subject,
                    data = dice_game_cleaned |>
                      filter(p_H == 1 | p_H == 0 | round == 4 |
                               round == 11  | round == 18),
                    family = binomial(link = "probit"))
reg3b_clustered_sd <- vcovCL(reg3b_probit, cluster = ~ subject)
mea_reg3b <- margins(reg3b_probit, variables = c("H", "L"),
                      vcov = reg3b_clustered_sd, atmeans = TRUE)

reg4b_probit <- glm(formula = add4 ~ H + L +
                      decision_time_task1_in_round +
                      decision_time_task2_in_round +
                      order_of_round + group_no + subject,
                    data = dice_game_cleaned |>
                      filter(p_H == 1 | p_H == 0 | round == 5 |
                               round == 12  | round == 19),
                    family = binomial(link = "probit"))
reg4b_clustered_sd <- vcovCL(reg4b_probit, cluster = ~ subject)
mea_reg4b <- margins(reg4b_probit, variables = c("H", "L"),
                      vcov = reg4b_clustered_sd, atmeans = TRUE)

reg5b_probit <- glm(formula = add4 ~ H + L +
                      decision_time_task1_in_round +
                      decision_time_task2_in_round +
                      order_of_round + group_no + subject,
                    data = dice_game_cleaned |>
```

Table 3: Table A.3: Probit Regression Results of Dice Game Experiment (Panel A. Full sample and subsamples by payoff pairs)

	All	All	40,p;0	30,p;0	22,p;18
H	0.150	0.149	0.175	0.232	0.035
	(0.026)	(0.025)	(0.037)	(0.034)	(0.036)
L	0.184	0.185	0.234	0.210	0.113
	(0.023)	(0.021)	(0.034)	(0.033)	(0.034)
Controls	N	Y	Y	Y	Y

```
                    filter(p_H == 1 | p_H == 0 | round == 6 |
                             round == 13  | round == 20),
                  family = binomial(link = "probit"))
reg5b_clustered_sd <- vcovCL(reg5b_probit, cluster = ~ subject)
mea_reg5b <- margins(reg5b_probit, variables = c("H", "L"),
                     vcov = reg5b_clustered_sd, atmeans = TRUE)

modelsummary(list(All = mea_reg1a, All = mea_reg2a,
                  '40,p;0' = mea_reg3a, '30,p;0' = mea_reg4a,
                  '22,p;18' = mea_reg5a),
             gof_map = c('nobs', 'pseudo.r.squared'),
             add_rows = data.frame(
               "term" = "Controls",
               "Model 1" = "N",
               "Model 2" = "Y",
               "Model 3" = "Y",
               "Model 4" = "Y",
               "Model 5" = "Y",
               check.names = FALSE
             ),
             title = 'Table A.3: Probit Regression Results of Dice Game Experiment (Panel A.
)
```

```
modelsummary(list('1/6' = mea_reg1b, '2/6' = mea_reg2b,
                  '3/6' = mea_reg3b, '4/6' = mea_reg4b,
                  '5/6' = mea_reg5b),
             gof_map = c('nobs', 'pseudo.r.squared'),
             add_rows = data.frame(
               "term" = "Controls",
```

Table 4: Table A.3: Probit Regression Results of Dice Game Experiment (Panel B. Subsamples by winning probabilities)

	1/6	2/6	3/6	4/6	5/6
H	0.123	0.173	0.126	0.131	0.154
	(0.025)	(0.028)	(0.029)	(0.031)	(0.026)
L	0.157	0.209	0.160	0.163	0.189
	(0.023)	(0.023)	(0.027)	(0.029)	(0.024)
Controls	Y	Y	Y	Y	Y

```
                "Model 1" = "Y",
                "Model 2" = "Y",
                "Model 3" = "Y",
                "Model 4" = "Y",
                "Model 5" = "Y",
                check.names = FALSE
              ),
              title = 'Table A.3: Probit Regression Results of Dice Game Experiment
              (Panel B. Subsamples by winning probabilities)'
)
```

The Appendix (Data Cleaning Code)

```
# get_cleaned_data_generalised (data cleaning function)
library(tidyverse)
library(haven)
library(purrr)
library(modelsummary)
library(xtable)
library(estimatr)
library(patchwork)
library(clubSandwich)
library(sandwich)
library(margins)

get_cleaned_data_generalised <- function(data1, outcome_variable, group_text_label,
                              ut, decision, decision_time_task1 = c(),
                              decision_time_task2 = c(), demographics,
                              belief_in_just_world = c(),
                              belief_in_prob = c(),
                              bonus_index) {
  data1 <- data1 |>
    select(any_of(c(ut, decision, decision_time_task1, decision_time_task2, demographics, bel
    mutate(
      across(
        .cols = all_of(ut),
        .fns = ~ ifelse(.x == bonus_index[cur_column()], 1, 0),
        .names = "ut_{col}")
    ) |>
    mutate(ut_sum = rowSums(across(starts_with('ut_'))),
           female = ifelse(!!sym(demographics[2]) == 2, 1, 0),
           age = 93 - !!sym(demographics[3]),
           econ = ifelse(!!sym(demographics[5]) == 2, 1, 0),
           no_religion = ifelse(!!sym(demographics[6]) == 7, 1, 0))
    if (length(belief_in_prob) != 0) {
      data1 <- data1 |>
        rename(belief_in_prob1 = !!sym(belief_in_prob[1]),
               belief_in_prob2 = !!sym(belief_in_prob[2]),
               belief_in_prob3 = !!sym(belief_in_prob[3]),
               belief_in_prob4 = !!sym(belief_in_prob[4]),
               belief_in_prob5 = !!sym(belief_in_prob[5])) |>
        mutate(belief_in_just_world4 = 8 - !!sym(belief_in_just_world[4]),
```

```
          belief_in_just_world = !!sym(belief_in_just_world[1]) + !!sym(belief_in_just_worl
  rename(belief_in_just_world1 = !!sym(belief_in_just_world[1]),
         belief_in_just_world2 = !!sym(belief_in_just_world[2]),
         belief_in_just_world3 = !!sym(belief_in_just_world[3]))
  } else {
  }


### Decisions

for (i in 1:21) {
  j <- i + 1
  data1[[paste0('box_receive_in_round', i)]] <- data1[[ decision[i] ]]
  if (length(decision_time_task1) != 0) {
    data1[[paste0('decision_time_task1_in_round', i)]] <- data1[[ decision_time_task1[i] ]]
  } else {
  }

  if (length(decision_time_task2) != 0) {
    data1[[paste0('decision_time_task2_in_round', i)]] <- data1[[ decision_time_task2[i] ]]
  } else {
  }
}

data1 <- data1 |>
  separate(!!sym(decision[length(decision)]), into = paste0(decision[length(decision)], 1:2

round_codes <- c(
  "JSZ","VXX","ECM","LNX","KUK","ITA","GVA",
  "HPK","VVA","VFX","CCJ","JBE","FEZ","TZT",
  "NMS","QZE","HMV","GXX","NXU","HNB","SFO"
)

for (i in 1:21) {
  data1[[paste0('round', i)]] <- round_codes[i]
}

fl_do_matrix <- as.matrix(data1 |>
                            select(starts_with(decision[length(decision)])))

for (i in 1:21) {
  target_round_code <- round_codes[i]
```

```
  order_values <- apply(fl_do_matrix, 1, function(row_values) {
    idx <- which(row_values == target_round_code)
    if (length(idx) == 0) {
      0
    } else {
      idx
    }
  })
  data1[[paste0('order_of_round', i)]] <- unlist(order_values)
}

data1 <- data1 |>
  select(-decision[decision != decision[length(decision)]],
         -decision_time_task1, -decision_time_task2,
         -starts_with(decision[length(decision)]))


data1_long <- data1 |>
  mutate(subject = row_number()) |>
    pivot_longer(
      cols = matches("^(box_receive_in_round|decision_time_task1_in_round|decision_time_tas
      names_pattern = "(.*?)(\\d+)$",
      names_to = c(".value", "round")
    ) |>
    mutate(round = as.integer(round))

data1_final <- data1_long |>
  # Map numeric round to Stata's round_code (JSZ, VXX, etc.)
  mutate(
    round_code = case_when(
      round == 1 ~ "JSZ",
      round == 2 ~ "VXX",
      round == 3 ~ "ECM",
      round == 4 ~ "LNX",
      round == 5 ~ "KUK",
      round == 6 ~ "ITA",
      round == 7 ~ "GVA",
      round == 8 ~ "HPK",
      round == 9 ~ "VVA",
      round == 10 ~ "VFX",
      round == 11 ~ "CCJ",
      round == 12 ~ "JBE",
```

```
      round == 13 ~ "FEZ",
      round == 14 ~ "TZT",
      round == 15 ~ "NMS",
      round == 16 ~ "QZE",
      round == 17 ~ "HMV",
      round == 18 ~ "GXX",
      round == 19 ~ "NXU",
      round == 20 ~ "HNB",
      round == 21 ~ "SFO",
      TRUE ~ NA_character_
    )
  ) |>
  # Calculate 'add4' based on round_code and box_receive_in_round
  mutate(
    !!sym(outcome_variable) := case_when(
      (round_code == "JSZ" & box_receive_in_round == 6) |
      (round_code == "VXX" & box_receive_in_round == 5) |
      (round_code == "ECM" & box_receive_in_round == 6) |
      (round_code == "LNX" & box_receive_in_round == 3) |
      (round_code == "KUK" & box_receive_in_round == 5) |
      (round_code == "ITA" & box_receive_in_round == 1) |
      (round_code == "GVA" & box_receive_in_round == 1) |
      (round_code == "HPK" & box_receive_in_round == 6) |
      (round_code == "VVA" & box_receive_in_round == 6) |
      (round_code == "VFX" & box_receive_in_round == 5) |
      (round_code == "CCJ" & box_receive_in_round == 5) |
      (round_code == "JBE" & box_receive_in_round == 3) |
      (round_code == "FEZ" & box_receive_in_round == 2) |
      (round_code == "TZT" & box_receive_in_round == 4) |
      (round_code == "NMS" & box_receive_in_round == 6) |
      (round_code == "QZE" & box_receive_in_round == 1) |
      (round_code == "HMV" & box_receive_in_round == 3) |
      (round_code == "GXX" & box_receive_in_round == 6) |
      (round_code == "NXU" & box_receive_in_round == 3) |
      (round_code == "HNB" & box_receive_in_round == 6) |
      (round_code == "SFO" & box_receive_in_round == 5) ~ 1,
      TRUE ~ 0
    )
  ) |>
  # Define experimental groups
  mutate(
    group = case_when(
```

```
        round >= 1 & round <= 7 ~ group_text_label[1],
        round >= 8 & round <= 14 ~ group_text_label[2],
        round >= 15 ~ group_text_label[3],
        TRUE ~ NA_character_
      ),
      group_no = case_when(
        round >= 1 & round <= 7 ~ 1,
        round >= 8 & round <= 14 ~ 2,
        round >= 15 ~ 3,
        TRUE ~ NA_integer_
      )
    ) |>
    # Define p_H (probability of High outcome)
    mutate(
      p_H = case_when(
        round %in% c(2, 9, 16) ~ 1/6,
        round %in% c(3, 10, 17) ~ 1/3,
        round %in% c(4, 11, 18) ~ 1/2,
        round %in% c(5, 12, 19) ~ 2/3,
        round %in% c(6, 13, 20) ~ 5/6,
        round %in% c(7, 14, 21) ~ 1,
        TRUE ~ 0 # Stata's default for unlisted rounds
      )
    ) |>
    select(
      subject, demographics, round, round_code, order_of_round,
      group, group_no, p_H, box_receive_in_round, female, age,
      starts_with('ut_'),
      econ, no_religion, !!sym(outcome_variable), ut,
      any_of(c('decision_time_task1_in_round',
               'decision_time_task2_in_round',
      starts_with('belief_in_just_world'), starts_with('belief_in_just_world'),
      starts_with('belief_in_prob')))
    ) |>
    rename(duration_in_seconds = Duration..in.seconds.)
}
```

```
# get 5 cleaned datasets


ut_dice <- c('Q1.2', 'Q1.5', 'Q1.8', 'Q1.11', 'Q1.14', 'Q1.17', 'Q1.20', 'Q1.23')
bonus_index_dice <- setNames(c(7, 3, 7, 7, 3, 1, 6, 3), ut_dice)
```

```
decision_dice <- c('Q2.6', 'Q3.6', 'Q4.6', 'Q5.6', 'Q6.6', 'Q7.6', 'Q8.6', 'Q9.6', 'Q10.6',
decision_time_task1_dice <- c('Q2.4_Page.Submit', 'Q3.4_Page.Submit', 'Q4.4_Page.Submit', 'Q5
decision_time_task2_dice <- c('Q2.8_Page.Submit', 'Q3.8_Page.Submit', 'Q4.8_Page.Submit', 'Q5
demographics_dice <- c('Duration..in.seconds.', 'Q23.2', 'Q23.3', 'Q23.4', 'Q23.5', 'Q23.6')
belief_in_just_world_dice <- c('Q23.7', 'Q23.8', 'Q23.9', 'Q23.10')
belief_in_prob_dice <- c('Q24.1_1', 'Q25.1_1', 'Q26.1_1', 'Q27.1_1', 'Q28.1_1')


# dice game
temp <- read.csv('C:\\Users\\75972\\Dropbox\\Further math\\ERM\\Replication Package\\Raw Data
group_text_label_temp <- c("40 vs. 0", "30 vs. 10", "22 vs. 18")

dice_game_cleaned <- get_cleaned_data_generalised(
  data1 = temp, outcome_variable = 'add4', group_text_label = group_text_label_temp,
  ut = ut_dice, decision = decision_dice,
  decision_time_task1 = decision_time_task1_dice,
  decision_time_task2 = decision_time_task2_dice, demographics = demographics_dice,
  belief_in_just_world = belief_in_just_world_dice,
  belief_in_prob = belief_in_prob_dice,
  bonus_index = bonus_index_dice)


# dice game loss
temp <- read.csv('C:\\Users\\75972\\Dropbox\\Further math\\ERM\\Replication Package\\Raw Data
group_text_label_temp <- c("44 vs. 4", "34 vs. 14", "26 vs. 22")
dice_game_loss_cleaned <- get_cleaned_data_generalised(
  data1 = temp, outcome_variable = 'deduct4', group_text_label = group_text_label_temp,
  ut = ut_dice, decision = decision_dice, decision_time_task1 = decision_time_task1_dice,
  decision_time_task2 = decision_time_task2_dice, demographics = demographics_dice,
  belief_in_just_world = belief_in_just_world_dice,
  belief_in_prob = belief_in_prob_dice,
  bonus_index = bonus_index_dice)


# direct choice
ut_direct <- c('Q1.2', 'Q1.5', 'Q1.8', 'Q1.11', 'Q1.14', 'Q1.17', 'Q1.20')
bonus_index_direct <- setNames(c(7, 3, 7, 7, 3, 1, 7), ut_direct)
decision_direct <- c('Q2.4', 'Q3.4', 'Q4.4', 'Q5.4', 'Q6.4', 'Q7.4', 'Q8.4',
                     'Q9.4', 'Q10.4', 'Q11.4', 'Q12.4', 'Q13.4', 'Q14.4',
                     'Q15.4', 'Q16.4', 'Q17.4', 'Q18.4', 'Q19.4', 'Q20.4', 'Q21.4', 'Q22.4',
decision_time_task2_direct <- c('Q2.6_Page.Submit', 'Q3.6_Page.Submit',
                                'Q4.6_Page.Submit', 'Q5.6_Page.Submit',
```

```
                                 'Q6.6_Page.Submit', 'Q7.6_Page.Submit',
                                 'Q8.6_Page.Submit', 'Q9.6_Page.Submit',
                                 'Q10.6_Page.Submit', 'Q11.6_Page.Submit',
                                 'Q12.6_Page.Submit', 'Q13.6_Page.Submit',
                                 'Q14.6_Page.Submit', 'Q15.6_Page.Submit',
                                 'Q16.6_Page.Submit', 'Q17.6_Page.Submit',
                                 'Q18.6_Page.Submit', 'Q19.6_Page.Submit',
                                 'Q20.6_Page.Submit', 'Q21.6_Page.Submit',
                                 'Q22.6_Page.Submit')
demographics_direct <- c('Duration..in.seconds.', 'Q23.2', 'Q23.3', 'Q23.4',
                         'Q23.5', 'Q23.6')
belief_in_just_world_direct <- c('Q23.7', 'Q23.8', 'Q23.9', 'Q23.10')
belief_in_prob_direct <- c('Q24.1_1', 'Q25.1_1', 'Q26.1_1', 'Q27.1_1', 'Q28.1_1')

temp <- read.csv('C:\\Users\\75972\\Dropbox\\Further math\\ERM\\Replication Package\\Raw Data
group_text_label_temp <- c("40 vs. 0", "30 vs. 10", "22 vs. 18")
direct_choice_cleaned <- get_cleaned_data_generalised(
  data1 = temp, outcome_variable = 'add4', group_text_label = group_text_label_temp,
  ut = ut_direct, decision = decision_direct,
  decision_time_task2 = decision_time_task2_direct, demographics = demographics_direct,
  belief_in_just_world = belief_in_just_world_direct,
  belief_in_prob = belief_in_prob_direct,
  bonus_index = bonus_index_direct)


# Ex Ante Resolution
ut_ex_ante <- c('Q1.2', 'Q1.5', 'Q1.8', 'Q1.11', 'Q1.14', 'Q1.17', 'Q1.20',
                'Q1.23')
bonus_index_ex_ante <- setNames(c(7, 3, 7, 7, 3, 1, 6, 3), ut_ex_ante)
decision_ex_ante <- c('Q2.5', 'Q3.5', 'Q4.5', 'Q5.5', 'Q6.5', 'Q7.5', 'Q8.5',
                      'Q9.5', 'Q10.5', 'Q11.5', 'Q12.5', 'Q13.5', 'Q14.5',
                      'Q15.5', 'Q16.5', 'Q17.5', 'Q18.5', 'Q19.5', 'Q20.5',
                      'Q21.5', 'Q22.5', 'FL_28_DO')
decision_time_task2_ex_ante <- c('Q2.7_Page.Submit', 'Q3.7_Page.Submit',
                                 'Q4.7_Page.Submit', 'Q5.7_Page.Submit',
                                 'Q6.7_Page.Submit', 'Q7.7_Page.Submit',
                                 'Q8.7_Page.Submit', 'Q9.7_Page.Submit',
                                 'Q10.7_Page.Submit', 'Q11.7_Page.Submit',
                                 'Q12.7_Page.Submit', 'Q13.7_Page.Submit',
                                 'Q14.7_Page.Submit', 'Q15.7_Page.Submit',
                                 'Q16.7_Page.Submit', 'Q17.7_Page.Submit',
                                 'Q18.7_Page.Submit', 'Q19.7_Page.Submit',
```

```
                                  'Q20.7_Page.Submit', 'Q21.7_Page.Submit',
                                  'Q22.7_Page.Submit')
demographics_ex_ante <- c('Duration..in.seconds.', 'Q23.2', 'Q23.3', 'Q23.4',
                          'Q23.5', 'Q23.7', 'Q23.4_6_TEXT','Q23.5_6_TEXT',
                          'Q23.6_14_TEXT', 'Q23.7_6_TEXT')
belief_in_just_world_ex_ante <- c('Q23.10_1','Q23.10_2', 'Q23.10_3', 'Q23.10_4')

temp <- read.csv('C:\\Users\\75972\\Dropbox\\Further math\\ERM\\Replication Package\\Raw Data
group_text_label_temp <- c("20 vs. 0", "15 vs. 5", "11 vs. 9")

ex_ante_resolution_cleaned <- get_cleaned_data_generalised(
  data1 = temp, outcome_variable = 'add2', group_text_label = group_text_label_temp,
  ut = ut_ex_ante, decision = decision_ex_ante,
  decision_time_task2 = decision_time_task2_ex_ante, demographics = demographics_ex_ante, bo

# Dictator Game
```

```
# load two second party datasets (cleaned dataset in the replication package)
second_party <- read_dta('C:\\Users\\75972\\Dropbox\\Further math\\ERM\\Replication Package\\
second_party_information <- read_dta('C:\\Users\\75972\\Dropbox\\Further math\\ERM\\Replicati

rename_duration <- function(data) {
  data |>
    rename(duration_in_seconds = durationinseconds)
}

data_list <- list(
  second_party = second_party,
  second_party_information = second_party_information
)
data_list <- map(data_list, rename_duration)
list2env(data_list, .GlobalEnv)
```

```
<environment: R_GlobalEnv>
```